

PHYSICS

Chapter: Waves

Types of Waves • Transverse & Longitudinal • Wavelength & Frequency • Practice Problems

Topics Covered	Key Formulas
▶ 1. What is a Wave? ▶ 2. Types of Waves ▶ 3. Transverse Waves ▶ 4. Longitudinal Waves ▶ 5. Wavelength (λ) ▶ 6. Frequency (f) and Period (T) ▶ 7. Wave Speed: $v = f\lambda$ ▶ 8. 10 Worked Examples ▶ 9. 10 Practice Problems	$v = f \times \lambda$ (wave speed) $f = v / \lambda$ (frequency) $\lambda = v / f$ (wavelength) $T = 1 / f$ (period) $f = 1 / T$ (frequency from T) $v = \lambda / T$ (speed from period) $c = 3 \times 10^8$ m/s (light in vacuum) $v_{\text{sound}} \approx 343$ m/s (air, 20°C)

Section 1: What is a Wave?

□ **Definition: Wave**

A wave is a disturbance that transfers energy from one place to another without permanently displacing the medium. The medium particles vibrate around their equilibrium positions but do not travel with the wave.

Key Properties of All Waves

Every wave — regardless of type — shares these fundamental properties:

Property	Symbol	Unit	Description
Wavelength	λ (lambda)	m (metres)	Distance between two consecutive identical points (e.g. crest to crest)
Frequency	f	Hz (hertz)	Number of complete oscillations per second
Period	T	s (seconds)	Time for one complete oscillation; $T = 1/f$
Amplitude	A	m (metres)	Maximum displacement from equilibrium position
Wave Speed	v	m/s	Distance travelled by the wave per second

Figure 1 — Classification of Waves

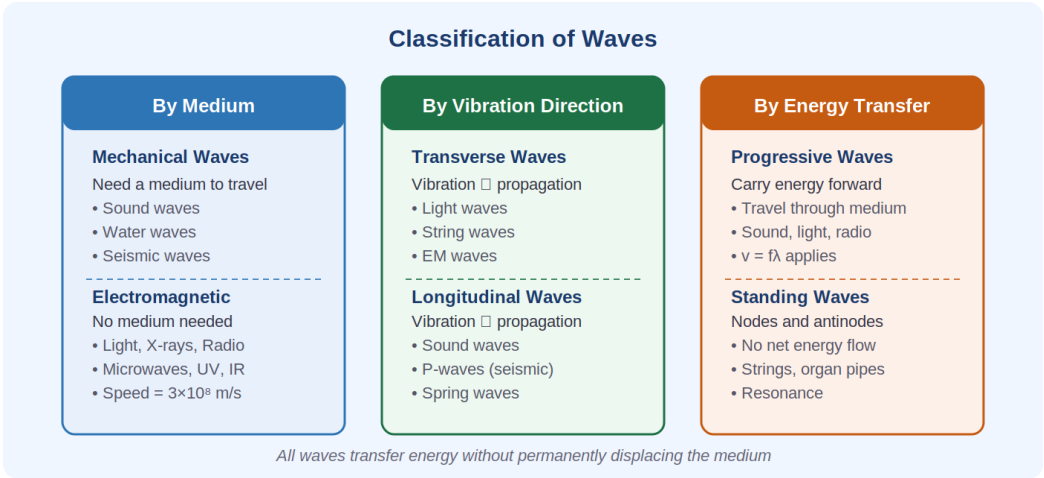


Figure 1: Three ways to classify waves — by medium required, by vibration direction, and by energy transfer mechanism.

□ **Note:** All waves obey the universal wave equation: $v = f \times \lambda$. This applies to sound, light, water waves, and every other type of wave.

Section 2: Transverse Waves

□ Definition: Transverse Wave

A transverse wave is one in which the particles of the medium vibrate perpendicular (at 90°) to the direction in which the wave travels. The displacement is at right angles to propagation.

Figure 2 — Transverse Wave Anatomy

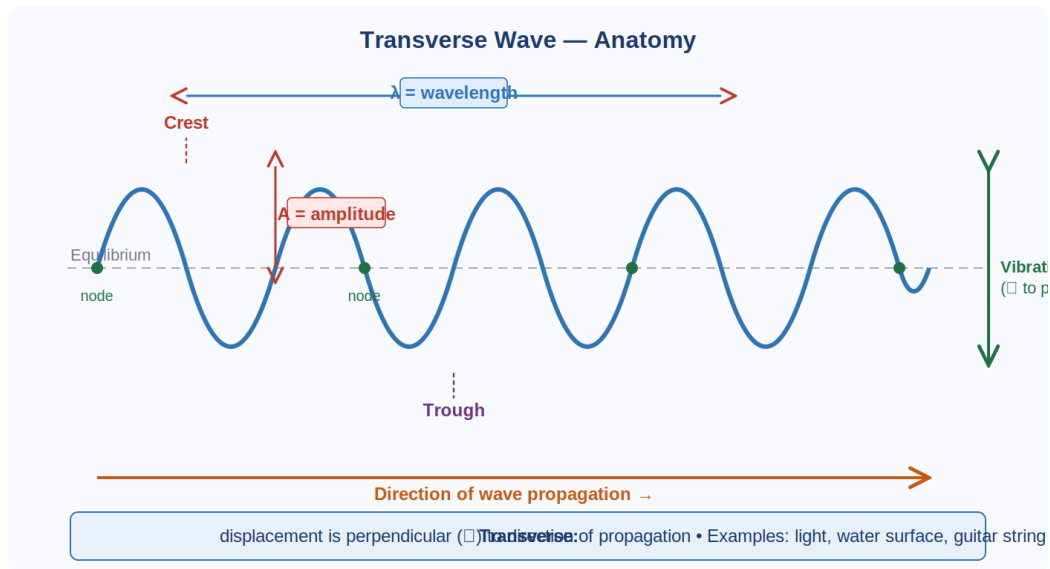


Figure 2: A transverse wave showing crests, troughs, amplitude (A), wavelength (λ), nodes, direction of vibration ($\updownarrow$), and direction of propagation ($\rightarrow$).

Parts of a Transverse Wave

Term	Definition	Diagram location
Crest	Highest point of the wave above equilibrium	Peak of sine curve
Trough	Lowest point of the wave below equilibrium	Valley of sine curve
Amplitude (A)	Maximum displacement from equilibrium	Crest height or trough depth
Wavelength (λ)	Distance from one crest to the next crest	Crest-to-crest distance
Node	Point where displacement is always zero	Crosses equilibrium line
Equilibrium	The undisturbed position of the medium	Centre line of the wave

Examples of Transverse Waves

The following are all transverse waves:

Light waves (electromagnetic)	All EM waves are transverse. Visible light, UV, X-rays, radio, microwaves.
Water surface waves	Surface of water moves up/down while wave travels horizontally.
Waves on a string	Plucking a guitar string — string moves up and down, wave travels along string.
S-waves (seismic)	Secondary seismic waves — particles move perpendicular to wave direction.

Section 3: Longitudinal Waves

□ Definition: Longitudinal Wave

A longitudinal wave is one in which the particles of the medium vibrate parallel to (in the same direction as) the direction of wave propagation. The wave consists of alternating compressions and rarefactions.

Figure 3 — Longitudinal Wave Anatomy

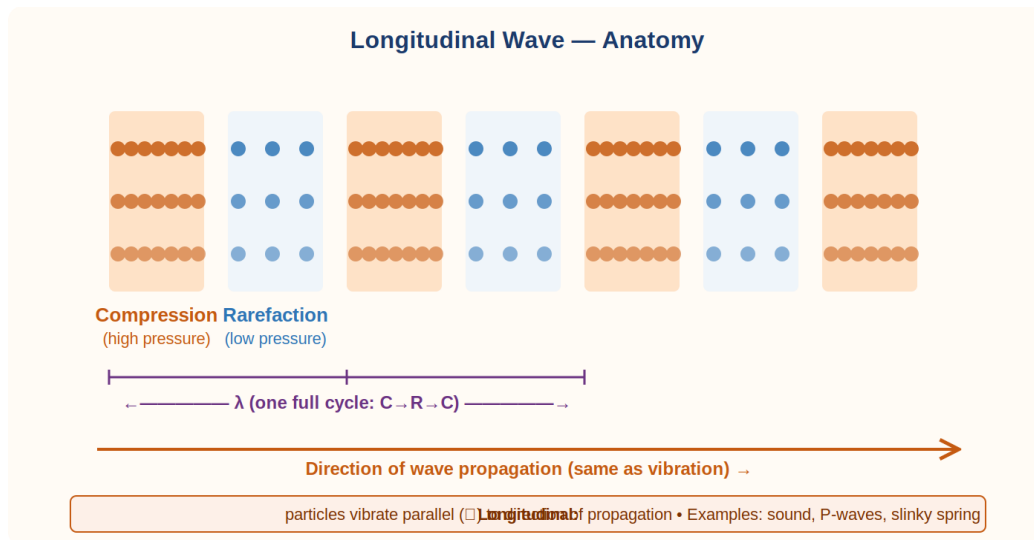


Figure 3: Longitudinal wave showing compression zones (orange — high density, high pressure) and rarefaction zones (blue — low density, low pressure). λ spans one full compression-rarefaction cycle.

Compressions and Rarefactions

Feature	Description	Analogous to (transverse)
Compression	Region where particles are pushed together; high pressure, high density	Crest
Rarefaction	Region where particles are spread apart; low pressure, low density	Trough
Wavelength	Distance from one compression centre to the next	Crest to crest
Amplitude	Maximum displacement of a particle from equilibrium	Same concept

Examples of Longitudinal Waves

The following are all longitudinal waves:

Sound waves	Pressure waves in air, water, or solids. Particles vibrate back-and-forth along propagation direction.
P-waves (seismic)	Primary seismic waves — fastest seismic wave, travels through solids and liquids.
Spring (slinky)	Push one end of a slinky — compressions travel along the spring's length.
Ultrasound	High-frequency sound above 20,000 Hz. Used in medical imaging and sonar.

Figure 4 — Side-by-Side Comparison

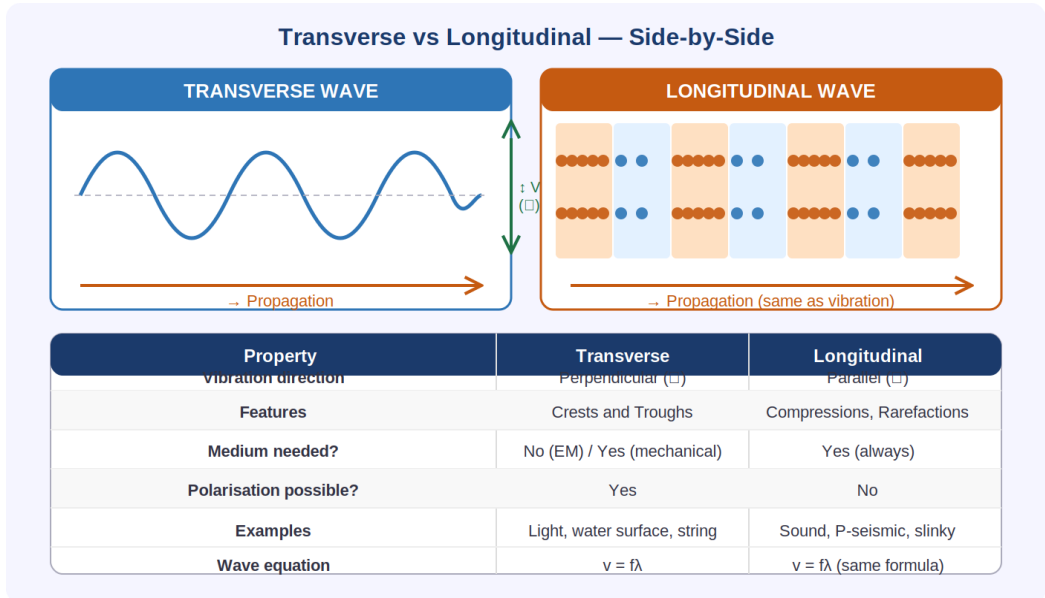


Figure 4: Transverse vs. Longitudinal comparison — waveform on top, property table below. Key difference: direction of particle vibration relative to wave travel.

Section 4: Wavelength, Frequency and Wave Speed

□ Definition: Wavelength (λ)

The wavelength is the distance between two consecutive points that are in phase — for example, crest to crest, or trough to trough, or compression to compression. Unit: metres (m).

□ Definition: Frequency (f)

Frequency is the number of complete wave cycles that pass a fixed point per second. Unit: hertz (Hz). 1 Hz = 1 cycle per second.

□ Definition: Period (T)

The period is the time taken for one complete oscillation (one full wave cycle). Period and frequency are inverses of each other: $T = 1/f$. Unit: seconds (s).

Figure 5 — Wavelength and Frequency

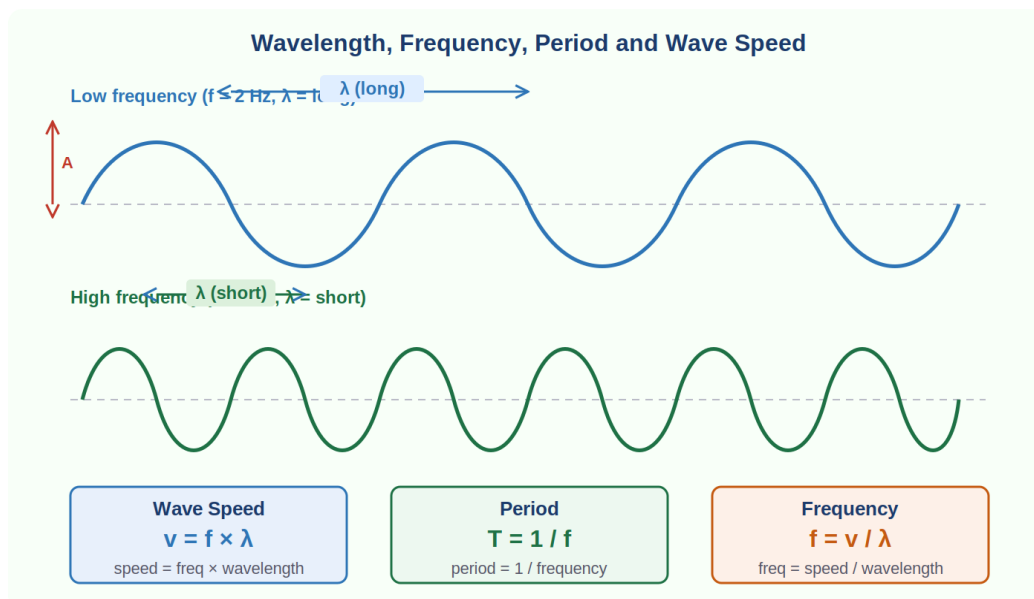


Figure 5: Two waves with same amplitude but different frequencies. Top: low frequency (long λ). Bottom: high frequency (short λ). Key equations shown in boxes.

The Wave Equation

The most important relationship in wave physics connects wave speed, frequency and wavelength:

Wave Speed Equation

$$v = f \times \lambda$$

Rearranged forms

$$f = v / \lambda$$

$$\lambda = v / f$$

$$T = 1 / f$$

Figure 6 — Formula Triangle

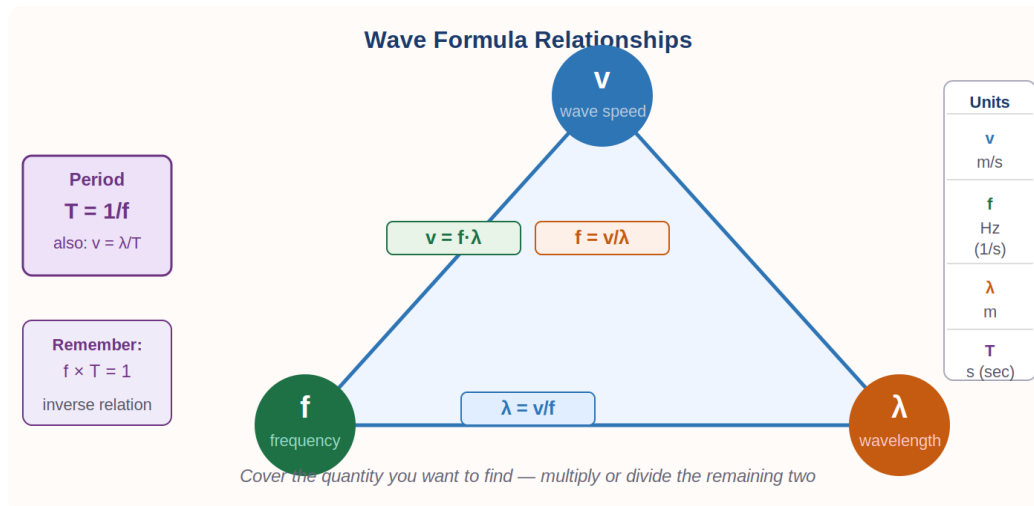


Figure 6: Formula triangle — cover the quantity you want to find; multiply or divide the remaining two. Units shown on right. Period relationship on left.

□ **Note:** When frequency increases, wavelength decreases (inversely proportional) — provided the wave speed stays constant. This is why high-pitched sounds have short wavelengths.

If f doubles	If λ doubles	If v doubles
λ halves (v constant)	f halves (v constant)	f doubles OR λ doubles
T halves, v unchanged	T doubles, v unchanged	f & λ multiply together $\rightarrow 2v$

Section 5: Worked Examples

Figure 7 — Wave Speed Reference

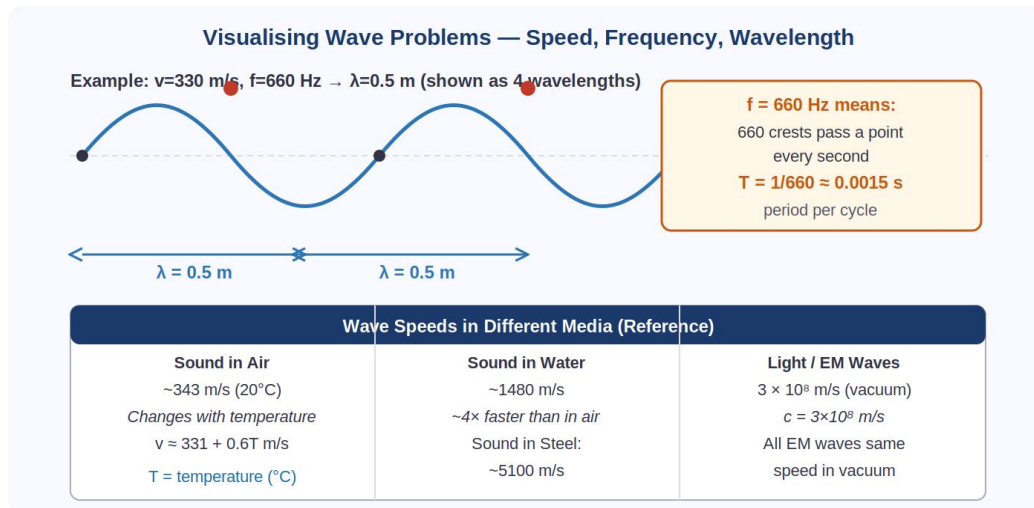


Figure 7: Visual showing wavelengths at 660 Hz and wave speed reference values in common media.

□ Example 1: Find wavelength given speed and frequency

Given: A sound wave travels through air at 340 m/s with a frequency of 680 Hz . Find its wavelength.

Step 1: Write down the formula: $v = f \times \lambda$

Step 2: Rearrange for λ : $\lambda = v / f$

Step 3: Substitute values:

$$\lambda = 340 / 680$$

$$\lambda = 0.5 \text{ m}$$

✓ **Answer:** $\lambda = 0.5 \text{ m}$ (50 cm) — about the length of a ruler

□ Example 2: Find frequency given speed and wavelength

Given: A water wave has a wavelength of 2.5 m and travels at 5 m/s . Find its frequency.

Step 1: Formula: $v = f \times \lambda \rightarrow$ rearrange to $f = v / \lambda$

Step 2: Substitute:

$$f = 5 / 2.5$$

$$f = 2 \text{ Hz}$$

✓ **Answer:** $f = 2 \text{ Hz}$ (2 complete waves pass a point every second)

□ **Example 3:** Find wave speed given frequency and wavelength

Given: A wave has frequency 250 Hz and wavelength 1.2 m. Find the wave speed.

Step 1: Use: $v = f \times \lambda$ (direct substitution)

Step 2: Substitute:

$$v = 250 \times 1.2$$

$$v = 300 \text{ m/s}$$

✓ **Answer:** $v = 300 \text{ m/s}$

□ **Example 4:** Find period then wavelength

Given: A wave has frequency 50 Hz and speed 200 m/s. Find (a) the period and (b) the wavelength.

Step 1: Period: $T = 1 / f$

$$T = 1 / 50 = 0.02 \text{ s}$$

Step 2: Wavelength: $\lambda = v / f$

$$\lambda = 200 / 50 = 4 \text{ m}$$

✓ **Answer:** (a) $T = 0.02 \text{ s}$ (b) $\lambda = 4 \text{ m}$

□ **Example 5:** Light wave — find frequency

Given: Yellow light has wavelength $580 \text{ nm} = 580 \times 10^{-9} \text{ m}$. Speed of light $c = 3 \times 10^8 \text{ m/s}$. Find its frequency.

Step 1: $f = v / \lambda = c / \lambda$

Step 2: Substitute:

$$f = (3 \times 10^8) / (580 \times 10^{-9})$$

$$f = (3 \times 10^8) / (5.8 \times 10^{-7})$$

$$f = 5.17 \times 10^{14} \text{ Hz} \approx 5.2 \times 10^{14} \text{ Hz}$$

✓ **Answer:** $f \approx 5.17 \times 10^{14} \text{ Hz}$ (517 trillion oscillations per second!)

□ **Example 6:** Find wavelength from period and speed

Given: A wave has a period of 0.004 s and travels at 320 m/s. Find the wavelength.

Step 1: Find frequency first: $f = 1 / T$

$$f = 1 / 0.004 = 250 \text{ Hz}$$

Step 2: Find wavelength: $\lambda = v / f$

$$\lambda = 320 / 250 = 1.28 \text{ m}$$

$$\text{Alternative: } \lambda = v \times T = 320 \times 0.004 = 1.28 \text{ m } \checkmark$$

✓ **Answer:** $\lambda = 1.28 \text{ m}$ (can also use $\lambda = v \times T$ directly)

□ **Example 7:** How many complete waves in a given distance?

Given: A sound wave has wavelength 0.8 m. How many complete wavelengths fit in a distance of 4.4 m?

Step 1: Number of wavelengths = distance / λ

$$n = 4.4 / 0.8$$

$$n = 5.5 \text{ wavelengths}$$

✓ **Answer:** 5.5 complete wavelengths

□ **Example 8:** Radio wave — find wavelength

Given: An FM radio station broadcasts at 99.5 MHz. Speed of EM waves = $3 \times 10^8 \text{ m/s}$. Find the wavelength.

Step 1: Convert frequency: $99.5 \text{ MHz} = 99.5 \times 10^6 \text{ Hz} = 9.95 \times 10^7 \text{ Hz}$

Step 2: $\lambda = v / f$

$$\lambda = (3 \times 10^8) / (9.95 \times 10^7)$$

$$\lambda = 3.015 \text{ m} \approx 3 \text{ m}$$

✓ **Answer:** $\lambda \approx 3 \text{ m}$ (that's why FM radio antennas are about 1–3 m long)

□ **Example 9:** Sound in water — find period

Given: Sound travels in water at 1480 m/s with wavelength 2.96 m. Find the frequency and period.

Step 1: Frequency: $f = v / \lambda$

$$f = 1480 / 2.96 = 500 \text{ Hz}$$

Step 2: Period: $T = 1 / f$

$$T = 1 / 500 = 0.002 \text{ s} = 2 \text{ ms}$$

✓ **Answer:** $f = 500 \text{ Hz}$, $T = 0.002 \text{ s}$ (2 milliseconds)

□ **Example 10:** Identify transverse or longitudinal + classify

Problem: Classify each wave as transverse (T) or longitudinal (L), and give one property:

(a) Sound in air →

L — Longitudinal. Particles vibrate parallel to wave direction. Creates compressions and rarefactions.

(b) Light from the Sun →

T — Transverse. Electromagnetic wave. Can be polarised. No medium needed. Speed = $3 \times 10^8 \text{ m/s}$ in vacuum.

(c) Waves on a guitar string →

T — Transverse. String vibrates perpendicular to wave direction. Demonstrates crests and troughs.

(d) Earthquake P-waves →

L — Longitudinal. Travel fastest of all seismic waves. Can pass through solid and liquid layers.

(e) Microwaves in oven →

T — Transverse. Electromagnetic wave, $f \approx 2.45 \text{ GHz}$, $\lambda \approx 12 \text{ cm}$. Absorbed by water molecules.

✓ **Answer:** (a) L (b) T (c) T (d) L (e) T

Section 6: Practice Problems

Instructions: Solve each problem showing all steps. Use $v = f\lambda$, $T = 1/f$. Answers are given at the end.

□ Practice 1: Basic wavelength calculation

A sound wave in air has a frequency of 440 Hz (note A4 on a piano). If the speed of sound in air is 343 m/s, calculate:

- (a) The wavelength of this sound wave
- (b) The period of this wave
- (c) How many complete wavelengths fit between a piano and a wall 10 m away?

□ Practice 2: Wave speed from frequency and λ

Ocean waves on a beach have a wavelength of 12 m and a period of 6 seconds.

- (a) Calculate the frequency of the waves
- (b) Calculate the speed of the waves
- (c) How far does the wave travel in 30 seconds?

□ Practice 3: High-frequency electromagnetic waves

A microwave oven operates at a frequency of 2.45 GHz = 2.45×10^9 Hz.

Speed of light $c = 3 \times 10^8$ m/s. Calculate:

- (a) The wavelength of the microwaves
- (b) The period of one oscillation
- (c) Is this wave transverse or longitudinal? Explain.

□ Practice 4: Finding frequency from period and speed

A vibrating string produces waves with a period of 0.005 s and wave speed of 80 m/s. Find:

- (a) The frequency of the wave
- (b) The wavelength of the wave
- (c) How many complete waves are produced in 3 seconds?

□ Practice 5: Comparing two waves

Wave A: frequency = 200 Hz, wavelength = 1.5 m

Wave B: frequency = 600 Hz, wavelength = 0.5 m

- (a) Calculate the wave speed of each wave
- (b) What can you conclude about both waves?
- (c) Which has the longer period? Calculate both.

□ **Practice 6:** Sound vs light timing

During a thunderstorm you see lightning and then hear thunder 3 seconds later.

Speed of sound = 343 m/s. Speed of light $\approx 3 \times 10^8$ m/s (instantaneous for these distances).

- (a) How far away is the lightning? (in metres and km)
- (b) If thunder has frequency 100 Hz, what is the wavelength of the sound?
- (c) Is thunder a transverse or longitudinal wave?

□ **Practice 7:** Ultrasound (medical imaging)

Medical ultrasound has a frequency of 5 MHz = 5×10^6 Hz.

Speed of ultrasound in soft tissue = 1540 m/s. Calculate:

- (a) Wavelength in tissue
- (b) Period of one oscillation
- (c) How long does it take ultrasound to travel 10 cm into tissue and reflect back (20 cm total)?

□ **Practice 8:** Identify wave type and properties

For each situation below, state: (i) transverse or longitudinal, (ii) whether it needs a medium:

- (a) A person speaking — sound travels through air to a listener
- (b) Sunlight reaching Earth through space
- (c) Sonar pulse from a submarine through water
- (d) A wave on a skipping rope
- (e) X-rays used in a hospital

□ **Practice 9:** Colour and wavelength (extension)

Visible light has wavelengths from 380 nm (violet) to 700 nm (red). $1 \text{ nm} = 1 \times 10^{-9} \text{ m}$.

Speed of light = 3×10^8 m/s. Calculate the frequency range of visible light:

- (a) Frequency of violet light ($\lambda = 380 \text{ nm}$)

(b) Frequency of red light ($\lambda = 700 \text{ nm}$)

(c) Which colour has the higher frequency? Higher energy?

□ **Practice 10:** Multi-step challenge problem

A sonar system sends a sound pulse into the ocean and receives an echo 2.4 seconds later. Speed of sound in seawater = 1500 m/s. The sonar frequency is 25 kHz = 25,000 Hz.

(a) How deep is the ocean floor at that point? (Remember: pulse travels there AND back)

(b) What is the wavelength of the sonar wave?

(c) What is the period of one sonar oscillation?

(d) Is sonar a transverse or longitudinal wave? Explain.

Section 7: Answers to Practice Problems

Q	Answer
P1	(a) $\lambda = 343/440 = 0.780 \text{ m}$ (b) $T = 1/440 = 0.00227 \text{ s}$ (c) $n = 10/0.780 = 12.8$ wavelengths
P2	(a) $f = 1/6 = 0.167 \text{ Hz}$ (b) $v = 0.167 \times 12 = 2 \text{ m/s}$ (c) $d = 2 \times 30 = 60 \text{ m}$
P3	(a) $\lambda = 3 \times 10^8 / 2.45 \times 10^9 = 0.122 \text{ m}$ (12.2 cm) (b) $T = 1/2.45 \times 10^9 = 4.1 \times 10^{-10} \text{ s}$ (c) Transverse – it is an EM wave
P4	(a) $f = 1/0.005 = 200 \text{ Hz}$ (b) $\lambda = 80/200 = 0.4 \text{ m}$ (c) $200 \times 3 = 600$ waves
P5	(a) Wave A: $v = 200 \times 1.5 = 300 \text{ m/s}$; Wave B: $v = 600 \times 0.5 = 300 \text{ m/s}$ (b) Same medium/type – same speed (c) $T_A = 1/200 = 5 \times 10^{-3} \text{ s}$; $T_B = 1/600 = 1.67 \times 10^{-3} \text{ s}$; Wave A has longer period
P6	(a) $d = 343 \times 3 = 1029 \text{ m} \approx 1.03 \text{ km}$ (b) $\lambda = 343/100 = 3.43 \text{ m}$ (c) Longitudinal
P7	(a) $\lambda = 1540/(5 \times 10^6) = 3.08 \times 10^{-4} \text{ m} = 0.308 \text{ mm}$ (b) $T = 1/(5 \times 10^6) = 2 \times 10^{-7} \text{ s}$ (c) $t = 0.20/1540 = 1.3 \times 10^{-4} \text{ s}$ (0.13 ms)
P8	(a) Longitudinal, needs medium (b) Transverse, no medium needed (c) Longitudinal, needs medium (d) Transverse, needs medium (e) Transverse, no medium needed
P9	(a) $f_{\text{violet}} = 3 \times 10^8 / 380 \times 10^{-9} = 7.89 \times 10^{14} \text{ Hz}$ (b) $f_{\text{red}} = 3 \times 10^8 / 700 \times 10^{-9} = 4.29 \times 10^{14} \text{ Hz}$ (c) Violet has higher frequency and higher energy ($E = hf$)
P10	(a) $d = (1500 \times 2.4) / 2 = 1800 \text{ m}$ (b) $\lambda = 1500/25000 = 0.06 \text{ m}$ (c) $T = 1/25000 = 4 \times 10^{-5} \text{ s}$ (d) Longitudinal – sound is always longitudinal

□ Quick Reference — Wave Equations

Wave Speed: $v = f \times \lambda$

(m/s = Hz $\times$ m)

Frequency: $f = v / \lambda$

(Hz = m/s $\div$ m)

Wavelength: $\lambda = v / f$

(m = m/s $\div$ Hz)

Period: $T = 1 / f$

(s = 1 / Hz)

From period: $v = \lambda / T$ and $f = 1 / T$

Light in vacuum: $c = 3 \times 10^8 \text{ m/s}$ (all EM waves)

Sound in air: $v \approx 343 \text{ m/s}$ at 20°C ($v \approx 331 + 0.6T$)
